

Department of Electronic & Telecommunication Engineering, University of Moratuwa, Sri Lanka.

Assignment 1

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Submitted in partial fulfillment of the requirements for the module

EN3330 Introduction to Engineering Optimization

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EN3330-Introduction to Engineering Optimization, Assignment 01

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(1.) (a.) $\min_{x} \quad \|x - v\|^2$

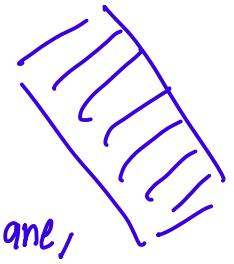
subject to $x^T u = d$

$\|u\|_2 = 1$ $\qquad x, v, u \in \mathbb{R}^n$



$x^T u = d$; $(n-1)$ dimensional hyper plane

Feasible region is the above mentioned hyper plane.



If both $x_1, x_2 \in \mathbb{R}^n$ lies on the hyper plane,

$x_1^T u = d$ — ①

$x_2^T u = d$ — ②

① − ② $\quad (x_1^T - x_2^T)\, u = 0$

$(x_1 - x_2)^T u = 0$

u is perpendicular to the hyper plane.

Optimal solution is when $(\vec{x} - \vec{v})$ is perpendicular to the hyper plane.

Therefore x^* is the point that is on the hyperplane which lies perpendicular to v.

$$x^* - v = \lambda u$$

$$x^* = v + \lambda u$$

should satisfy $x^T u = d$

$$[v + \lambda u]^T u = d$$

$$v^T u + \lambda \|u\|^2 = d$$

$$\lambda = d - v^T u$$

optimum solution; $x^*_1 = v + (d - v^T u)\, u$ //

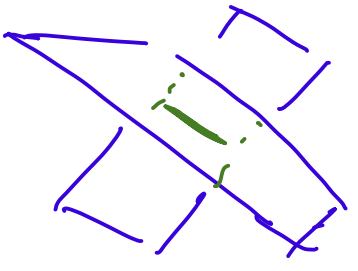
At the optimum,

$$(\vec{v} - \vec{x^*}) \;//\; \vec{u}$$

//

(b.) constraints: $x^T u_1 = d_1$, $x^T u_2 = d_2$

Now the feasible region is the intersection of both the hyperplanes.



Therefore x^* is the closest point to v in the intersection of the above two hyperplanes.

$x^* - v$ is also perpendicular to the intersection line.

Both u_1 & u_2 are also $\perp$ to the $\;\;$.

$$x^* - v = \lambda_1 u_1 + \lambda_2 u_2$$

Should satisfy both the below equations.

$$x^T u_1 = d_1$$

$$(v + d_1 u_1 + \lambda_2 u_2)^T u_1 = d_1$$

$$v^T u_1 + \lambda_1 + \lambda_2 u_2^T . u_1 = d_1$$

$$d_1 + \lambda_2 u_2^T u_1 = d_1 - v^T u_1 \quad \text{— (1)}$$

$$x^T u_2 = d_2$$

$$(v + d_1 u_1 + d_2 u_2)^T u_2 = d_2$$

$$v^T u_2 + d_1 u_1^T u_2 + d_2 = d_2$$

$$\lambda_1 u_1^T u_2 + \lambda_2 = d_2 - v^T u_2 \quad \text{— (2)}$$

$u_1^T u_2 = a$ $\quad d_2 - v^T u_2 = b$ $\quad u_2^T u_1 = c$ $\quad d_1 - v^T u_1 = d$

$$\lambda_1 + \lambda_2 c = d \quad \text{— (1')}$$

$$d_1 a + d_2 = b \quad \text{— (2')}$$

using cramers rule,

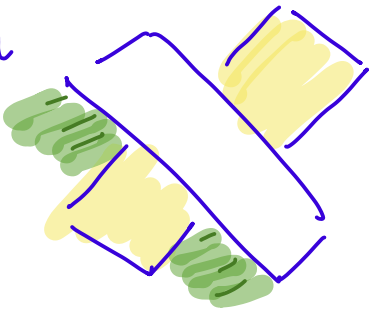
$$\lambda_1 = \frac{\begin{vmatrix} 1 & c \\ a & 1 \end{vmatrix}}{\begin{vmatrix} d & c \\ b & 1 \end{vmatrix}} = \frac{1 - ca}{d - bc}$$

optimal solution:

$$x_2^* = v + \lambda_1 u_1 + \lambda_2 u_2 \; //$$

$$\lambda_2 = \frac{\begin{vmatrix} 1 & c \\ a & 1 \end{vmatrix}}{\begin{vmatrix} 1 & d \\ a & b \end{vmatrix}} = \frac{1 - ca}{b - ad}$$

(C.) constraints : $x^T u_1 = d_1$ $x^T u_2 \geq d_2$



optimum solution should satisfy both the constraints.

case 1

If $x^T u_1 \geq d_2$ then it's already in the feasible region then the inequality is inactive.

$x^* = x_1^*$

$x^* = v + (d - v^T u)\, u$

case 2

If projection of v is in the opposite side of the feasible region then to satisfy both the above constraints x^* should be on the intersection that is

$x^T u_1 = d_1$ $x^T u_2 = d_2$.

$x^* = x_2^* = v + \lambda_1 u_1 + \lambda_2 u_2$ //

(2) (a) $b = \phi\, w$

$$b = \begin{bmatrix} y_1 \\ \vdots \\ y_p \end{bmatrix} \qquad w = \begin{bmatrix} x_1 \\ \vdots \\ x_q \end{bmatrix}$$

$$\phi = \underbrace{\begin{bmatrix} | & | & | \end{bmatrix}}_{q} \Big\}\, p$$

(a) When $p < q$ $\quad q$ = No of unknowns

Rank of ϕ = min (p, q)

Since $p < q$

rank of $\phi = p$

Since $p < q$

There are infinitely many solutions.

When $p > q$

rank of $\phi = q$ = # of unknowns

Since $p > q$, more equations than unknown variables.

Therefore No exact solution.

When $P = Q$

rank of Φ = $Q = P$ = # of unknowns

= No of equations

There is a unique solution.

(b.) l_2 -norm normalization

$$\arg\min_{\omega} \|\omega\|_2^2 \quad \text{subject to } b = \Phi\omega$$

l_∞ - norm normalization.

$$\arg\min_{\omega} \|\omega\|_\infty \quad \text{subject to } b = \Phi\omega$$

(c.) $P = 1 \quad Q = 2$

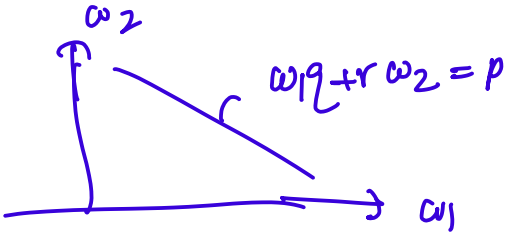
$b = \Phi\omega \qquad b = [p] \qquad \Phi = [q \quad r]$

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

$$\Phi\omega = b$$

$$[q \quad r] \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} = [p]$$

$q\omega_1 + r\omega_2 = p$; straight line



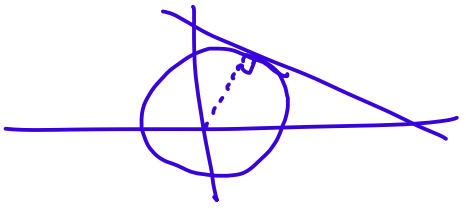
Option 1

l_2 norm minimization

minimize $\|w\|_2^2 = w_1^2 + w_2^2$

level sets $w_1^2 + w_2^2 = k^2$; circle

Point of intersection is the projection of the origin to the line. And is perpendicular



Option 2

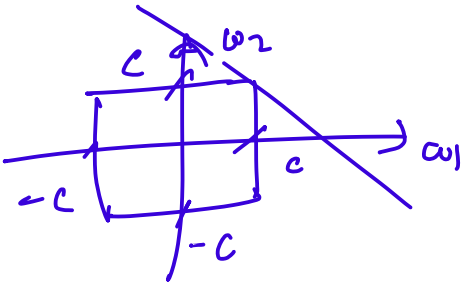
l_∞ norm minimization

$l_\infty = \max(|w_1|, |w_2|)$

suppose $l_\infty = c$

$-c \leq w_1 \leq c$

$-c \leq w_2 \leq c$



point of intersection is the corner of the square.

(d.) l_∞ cares only about the max value among $w \in \mathbb{R}^9$

In l_∞ level sets are squares.

the smallest square touching the solution line touches it at a corner. At a corner, $|\omega_1| = |\omega_2|$

Cohen generalized to $\mathbb{R}^q$ $|\omega_1| = |\omega_2| = \ldots = |\omega_q|$ //

In the solution all the components have equal absolute values.